# Appendix: On the Emergence of Multidimensional Structure from the Logarithmic Fundamental Quantity

## — A Formal Observation on the Dimension of the Hyperplane Defined by the Additive Identity

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## Purpose

The related paper stated that structure arises under the strong constraints of high symmetry and anonymity. In the main text, however, it did not exhibit that "arising structure" in a concrete form. This appendix is a **formal observation** intended to fill that gap.

Namely, from the minimal starting point of the fundamental quantity $N = \log_2 \nu$ and the additive identity of the conjugate relation alone, increasing the number of variables makes a multidimensional hyperplane structure arise formally — this is shown, as an example, by increasing the number of variables up to four.

**What this appendix does not claim (stated up front):**

- That this connects to physical spacetime / dimensions. Not claimed at all.
- That four dimensions are special, or that things stabilize at four. Not claimed. Four is merely an example and is given no further meaning. Asking why a particular dimension would rapidly make the discussion difficult, so this appendix does not enter there.
- That this is a mechanism of dimensional emergence. Not claimed.

This appendix is confined to an illustration of structure: "from the minimal logarithmic structure, a multidimensional hyperplane can be formally derived according to the number of variables."

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## 1. Starting point: the fundamental quantity N (one variable)

As in §4 of the related paper, the fundamental quantity is introduced as the logarithm of a squared quantity:

$$N \equiv \tfrac{1}{2}\log_2(\nu^2) = \log_2 \nu.$$

When $\nu = 2^n$, N=n (integer). N appears as a single axis (the whole integer set $\mathbb{Z}$). This is, on the face of it, a one-dimensional structure.

The question of this appendix is: when this fundamental quantity is placed in several copies (n of them), does a higher-dimensional hyperplane structure arise formally according to the number of variables? The multidimensionality does not "spring from nothing out of a single axis"; it comes from the number n of variables placed — this is stated up front.

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## 2. The hyperplane defined by the additive identity

As in §5 of the related paper, taking the base-2 logarithm of the conjugate relation $\nu\lambda = k$ turns the product into a sum:

$$\nu\lambda = k \xrightarrow{\log_2} N_\nu + N_\lambda = K \quad (K \equiv \log_2 k).$$

We now generalize. Take n fundamental quantities $N_1, N_2, \ldots, N_n$ and require their sum to be held at a constant value K:

$$\sum_{i=1}^{n} N_i = K.$$

This is a single linear equation in n variables, and it defines an **(n − 1)-dimensional hyperplane** within the n-dimensional coordinate space $(N_1, \ldots, N_n)$.

That is, the additive identity raises a flat structure (a hyperplane) of dimension (n − 1) according to the number of variables n. Below we look concretely at n=2, 3, 4.

**Positioning within the axiom system (important):** Placing several variables ($n \geq 3$) is itself an operation beyond Axiom 2 of the related paper ("only up to 2 can be distinguished in the interior"), and **presupposes the dimensional generation of §3** (the genesis of an integer mode number n via boundedness = the breaking of anonymity). That is, this appendix shows that, in the regime after anonymity has been broken and dimension has arisen, the additive identity defines

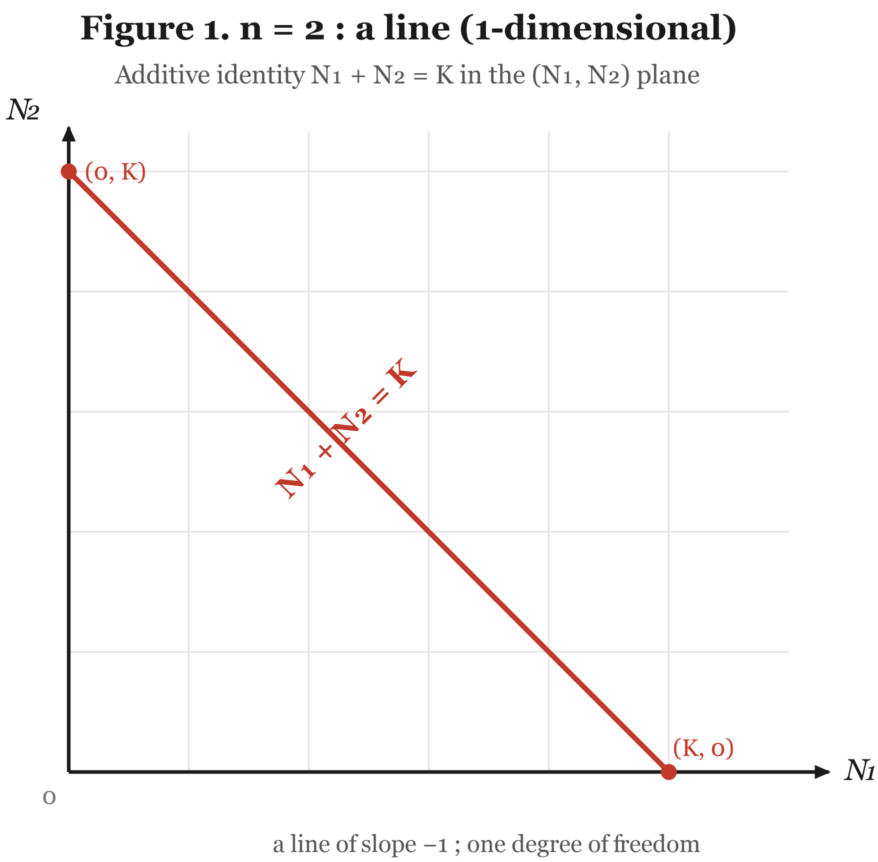


図 1　Figure 1: n=2. The line of slope − 1 defined by the additive identity $N_1+N_2=K$ (1-dimensional)

an (n − 1)-dimensional hyperplane. §3 (the genesis of dimension) and this appendix (n variables → an (n − 1)-dimensional hyperplane) stand in an upstream/downstream relation.

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## 3. n=2: a line (one-dimensional)

The sum of two variables $N_1, N_2$ is constant:

$$N_1 + N_2 = K.$$

This is a **line** of slope − 1 in the $(N_1, N_2)$ plane (one-dimensional). The relation "the sum of the conjugate pair is constant" appears as a single line in the plane (Figure 1).

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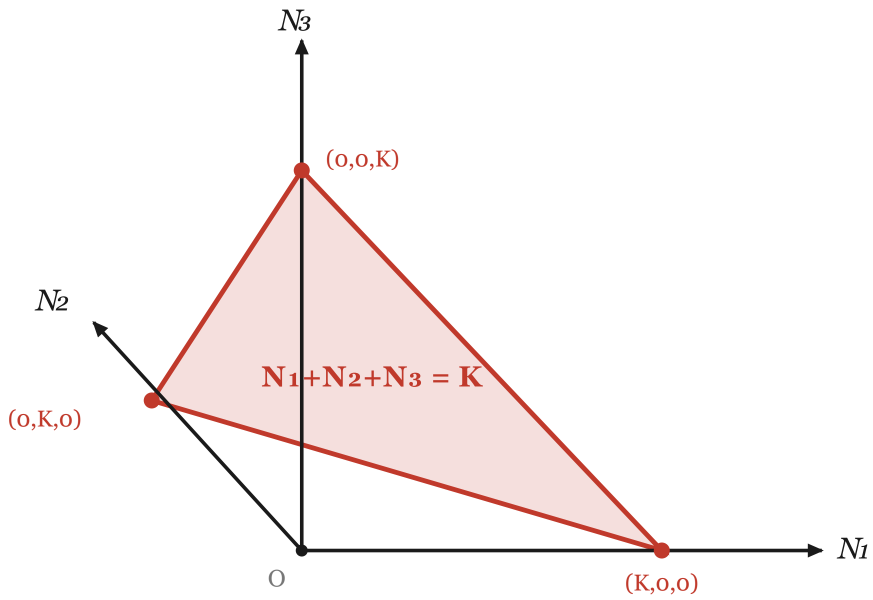
**Figure 2. n = 3 : a plane (2-dimensional)**

Additive identity N1 + N2 + N3 = K in (N1, N2, N3) space




a planar triangle in the positive octant ; two degrees of freedom

図 2 Figure 2: n=3. The plane defined by the additive identity N1+N2+N3=K (2-dimensional). The positive region is a triangle (2-simplex)

## 4. n=3: a plane (two-dimensional)

The sum of three variables $N_1, N_2, N_3$ is constant:

$$N_1 + N_2 + N_3 = K.$$

This is a **plane** in the three-dimensional coordinate space $(N_1, N_2, N_3)$ (two-dimensional). In the positive region of the three axes, this plane appears as a triangular region (Figure 2).

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## 5. n=4: a 3-dimensional hyperplane (three-dimensional)

The sum of four variables $N_1, N_2, N_3, N_4$ is constant:

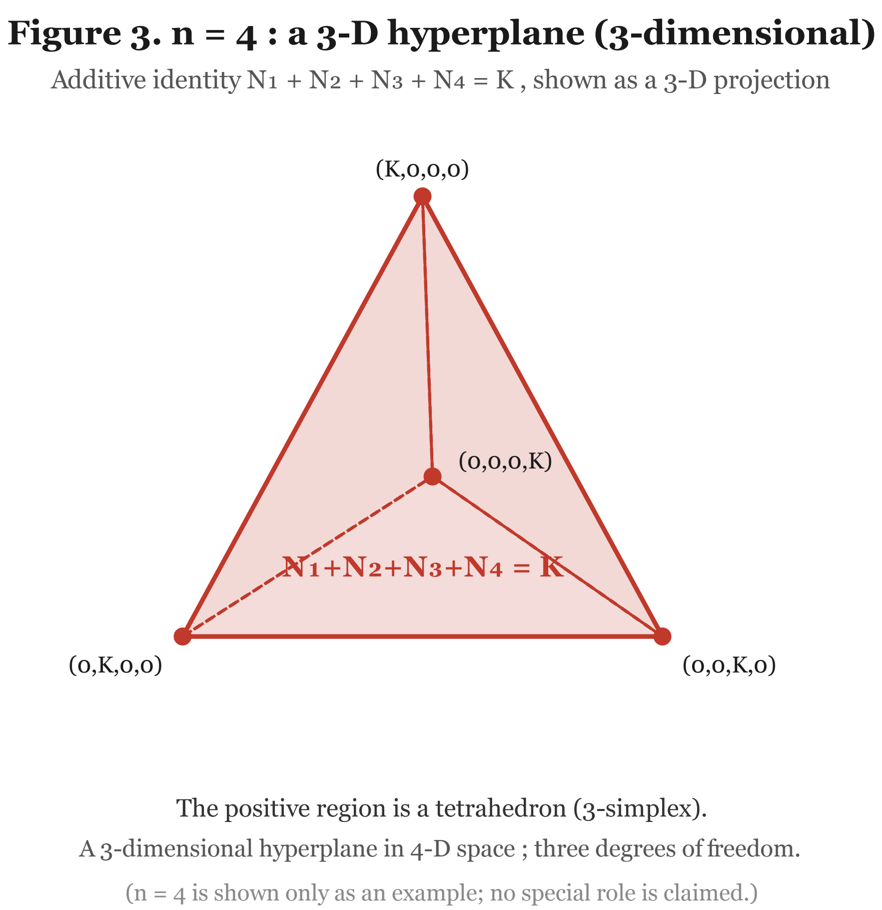


図 3 Figure 3: n=4. The 3-dimensional hyperplane defined by the additive identity $N_1+N_2+N_3+N_4=K$ (3-dimensional). The positive region is a tetrahedron (3-simplex); the vertices are the four axis intercepts

$$N_1 + N_2 + N_3 + N_4 = K.$$

This is a **3-dimensional hyperplane** in four-dimensional coordinate space. Since four dimensions cannot be drawn directly, it is shown as a projection into three dimensions. In the positive region of the four axes, this hyperplane appears as a tetrahedral (simplex) region (Figures 3 and 4).

Increasing the number of variables to four here is merely an example. No special meaning is given to the number four. Increasing the number of variables further raises an $(n - 1)$-dimensional hyperplane in the same manner.

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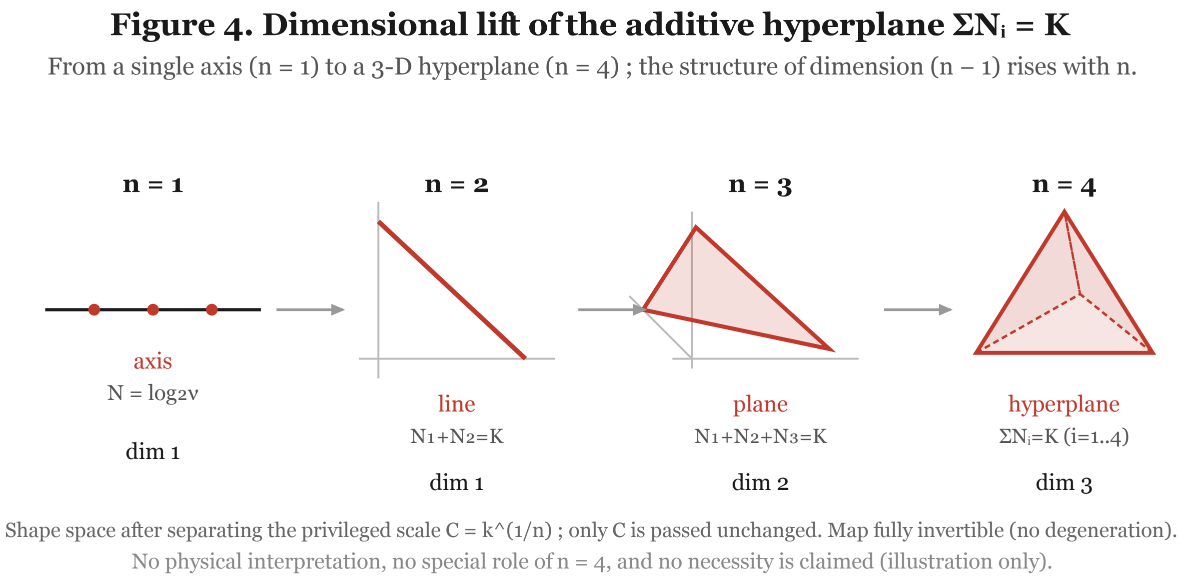


図 4 Figure 4: n=1 (axis) → n=2 (line) → n=3 (plane) → n=4 (tetrahedron). The dimension of the hyperplane defined by the additive identity $\Sigma\ N_i=K$ rises as (n − 1) with the number of variables n

## 6. Summary

From the minimal logarithmic fundamental quantity $N=\log_2\ \nu$ and the additive identity $\Sigma\ N_i=K$ alone, the following structure arises formally according to the number of variables n:

| Number of variables n | Additive identity | Arising structure | Dimension |
|---|---|---|---|
| 1 | (a single axis) | axis | 1 |
| 2 | $N_1+N_2=K$ | line | 1 |
| 3 | $N_1+N_2+N_3=K$ | plane | 2 |
| 4 | $N_1+N_2+N_3+N_4=K$ | 3-D hyperplane | 3 |
| n | $\Sigma\ N_i=K$ | hyperplane | n − 1 |

Placing the fundamental quantity $N=\log_2\ \nu$ in several copies (n of them) and, from the additive identity $\Sigma\ N_i=K$ that holds their sum constant alone, an (n − 1)-dimensional hyperplane arises formally according to the number n of variables. The multidimensionality does not spring from a single axis; it comes from the number of variables placed. This is the minimal concrete example of what the related paper meant by “structure arises.”

These hyperplanes are all flat (carry no curvature). This is consistent with the observation in §7 of the related paper that “the logarithmic map sends the additive identity $\Sigma\ N_i=K$ to a flat hyperplane,” and **it holds in any dimension without requiring full symmetry**.

**Canonicalization and the privileged scale (see the appendix “The maps (ν, δ) ↔ (N,**

**Δ)”):** separating the conserved quantity as a privileged scale C=k^{1/n}, the hyperplane $\Sigma N_i=K$ can be written as the normalized shape $\Sigma N_i=0$ ($N_i=\log_2(\nu_i/C)$). Only C is passed through unchanged (a privileged coordinate), and the map is fully invertible (the individual values and fluctuations are recovered exactly). Each figure shows the shape space after C has been separated; the scale C is to be read as a privileged coordinate attached separately to the figure.

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## 7. Limitations

1. **Four is merely an example.** Increasing the number of variables to four has no special meaning. No claim is made that four dimensions stabilize or are special. The appendix does not enter the question of why a particular dimension (entering it would rapidly make the discussion difficult).
2. **No physical connection is claimed.** It is not claimed that the arising hyperplane connects to physical spacetime / dimensions / geometry. This appendix is a purely formal illustration of structure and does not address isomorphism with a physical theory.
3. **This is an illustration of possibility, not a proof of necessity.** It only shows by one example that “a multidimensional structure can arise,” and does not show that it “must arise.”

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*This appendix is a formal illustration of the related paper’s claim (that structure arises under strong constraints), and contains no definitive claim regarding physical connection, the necessity of a particular dimension, or stability.*